

Project work investment banking

Explanation of the main models

1. ARIMA (1,0,0) model

The ARIMA (AutoRegressive Integrated Moving Average) model is a class of models used to describe the dynamics of the mean of a time series. In the specific case of ARIMA (1,0,0), the model reduces to a first-order autoregressive process, in which the current value of the series depends linearly on the value observed in the previous period.

In the context of financial time series, returns generally exhibit little or no autocorrelation, making the predictive capability of ARIMA models on the mean limited. However, the use of a low-order ARIMA makes it possible to remove any residual dependencies in the mean, preventing these components from contaminating the volatility analysis.

In the present work, the ARIMA (1,0,0) model is not used as a tool for forecasting returns, but rather as a proxy for volatility.

2. GARCH (1,1) model

The GARCH (Generalized Autoregressive Conditional Heteroskedasticity) model is designed to model the conditional volatility of financial time series, which are characterized by heteroskedasticity and volatility clustering. In such series, variance is not constant over time but depends on past information.

The GARCH (1,1) is widely used in finance because it allows for obtaining a dynamic and forward-looking measure of risk, making it particularly suitable for risk management and portfolio optimization contexts.

3. Use of ARIMA as a fallback for GARCH

Despite its theoretical effectiveness, the GARCH model may present convergence problems or parameter instability, especially in short time windows or during periods of low volatility. In such cases, the estimation of conditional volatility may prove unreliable or unavailable.

To overcome this limitation, in the present work a fallback approach based on ARIMA (1,0,0) has been adopted. In particular, when the GARCH model is not applicable, volatility is estimated as the variance of the residuals of the ARIMA model.

In fact, it is not used as a structural model of volatility, but as a fallback tool to obtain a reliable estimate of variance when GARCH is not applicable. In this way, stability of the estimate is favored over dynamics, consistently with the objective of portfolio construction.

NB All performance measures are calculated on a weekly basis, in line with the portfolio rebalancing frequency. This choice avoids annualizations that would introduce additional assumptions on the distribution and independence of returns, allowing for a more transparent and methodologically consistent comparison among the different strategies analyzed.

Part 1 – Classification of stocks by risk class and comparison between forecasted and historical volatility

The objective of the first part of the work is to assess whether the use of a forecast-based measure of volatility, obtained through GARCH models, provides an informational advantage over a purely historical measure, in the context of constructing equity portfolios classified by risk profile.

Volatility represents one of the central variables in modern financial theory, both as a proxy for risk and as a fundamental input in asset allocation processes. However, while historical volatility reflects exclusively the past behavior of returns, GARCH models make it possible to estimate conditional volatility, capable of dynamically adapting to changes in market regimes and of capturing phenomena such as volatility clustering and conditional heteroskedasticity, widely documented in the empirical literature.

Backtest structure

The backtest is set up according to a walk-forward logic with weekly rebalancing. At each rebalancing date, only the information available up to that moment is used, avoiding any form of look-ahead bias. Moreover, the investable universe is defined dynamically using the historical composition of the S&P 500 index, in order to eliminate survivorship bias and ensure temporal consistency between data and investment opportunities.

For each stock belonging to the investable universe, volatility is estimated over a rolling window of 252 trading days using two distinct approaches:

- forecasted volatility, obtained through a GARCH (1,1) model;
- historical volatility, calculated as the standard deviation of returns over the same window.

Classification by risk classes

Once volatilities are estimated, stocks are classified into three risk classes – conservative, moderate, and aggressive – using the tertiles of the cross-sectional distribution of estimated volatility at the rebalancing date. This choice makes it possible to obtain a relative classification, adaptive to the current market context, avoiding arbitrary thresholds and ensuring a balanced size of the classes over time. The classification is performed separately for the two volatility measures, producing two distinct partitions of the investable universe: one based on GARCH volatility and one on historical volatility.

Portfolio construction and performance measurement

For each classification method, three equal-weight portfolios are constructed, one for each risk class, for a total of six strategies. The use of uniform weights is a deliberate choice, aimed at isolating the effect of stock classification and the adopted risk measure, without introducing additional assumptions related to optimization models or weight estimation.

The returns of each portfolio are calculated on a weekly basis, assuming a buy-and-hold strategy between two consecutive rebalancing dates. This makes it possible to obtain comparable return time series, on which cumulative performance, volatility, Sharpe ratio, maximum drawdown, and frequency of relative outperformance are subsequently analyzed.

Purpose of the analysis

The comparison between portfolios constructed based on forecasted and historical volatility makes it possible to assess whether the information contained in the conditional volatility estimated by GARCH models translates into an actual improvement in the risk–return characteristics of the portfolios, or whether, on the contrary, the high reactivity of the model during periods of market stress may introduce instability in classification and performance.

Commentary on the results – Part 1

Conservative class

Graphical comparison and cumulative performance

The graphical comparison between GARCH conservative and historical volatility conservative shows a much closer behavior than that observed in the moderate class. The two cumulative return curves are often overlapping, with a slight but persistent predominance of the portfolio based on historical volatility in the long run. In the invested capital chart (base 100), the historical strategy shows a slightly more regular trajectory and higher final growth. This suggests that, for low-risk stocks, historical volatility is able to capture the risk structure in a more stable manner.

Quantitative performance comparison

Performance metrics confirm what emerges from the visual analysis:

- Weekly average return: Historical (0.00180) > GARCH (0.00175)
- Sharpe Ratio: Historical (0.0596) > GARCH (0.0574)
- Final cumulative return: Historical (1.25) > GARCH (1.19)
- Max Drawdown: Historical (−0.3216) < GARCH (−0.3354)

The portfolio based on historical volatility therefore appears dominant across all main indicators, albeit with contained differences. From a risk–return perspective, the GARCH approach fails to generate sufficient excess return to compensate for the increase in instability introduced by the higher reactivity of the model.

Analysis of stress weeks

The analysis focused on stress weeks further reinforces this evidence. In the presence of market shocks:

- the GARCH conservative portfolio outperforms the historical counterpart only in 44.44% of cases. This behavior is consistent with the nature of GARCH models, which tend to react in an amplified way to volatility shocks. In a conservative context, such reactivity leads to frequent non-optimal reallocations, reducing the effectiveness of protection during the most turbulent market phases.

Analysis of the worst weekly drawdowns

The analysis of the five worst weekly crashes shows that the deepest losses of the GARCH portfolio are concentrated around the events of March–April 2020. In several of these weeks, the portfolio based on historical volatility records significantly smaller losses, and in some cases even positive returns. This highlights that, for the conservative class, the use of GARCH volatility may prove counterproductive during periods of systemic stress.

Comparison of portfolio composition

The composition comparison at the date 2019-11-15 shows a high overlap between the two portfolios. This result indicates that, in the conservative class, the two strategies produce very similar classifications. However, the small differences introduced by the GARCH model do not translate into greater instability during stress phases.

Overall assessment

Overall, the results for the conservative class show that:

- historical volatility provides a more stable and effective classification for low risk stocks;
- the greater reactivity of the GARCH model proves detrimental in a context in which capital preservation is the priority. These findings suggest that the use of GARCH volatility alone is less appropriate for strategies with a conservative orientation.

Moderate class

From the graphical comparison between GARCH moderate and historical volatility moderate, a clear and persistent dynamic over time emerges. The portfolio based on GARCH volatility shows a systematically higher cumulative return compared to the counterpart constructed using historical volatility, particularly in the period after 2020.

The invested capital chart (base 100) highlights how the GARCH strategy is able to capture market expansion phases more effectively, accumulating an advantage that consolidates over the long term. Conversely, the portfolio based on historical volatility appears slower in adapting to changes in the risk regime, with a less pronounced growth trajectory.

Performance comparison

Performance metrics confirm what is observed visually:

- Weekly average return: GARCH (0.00240) > Historical (0.00225)
- Sharpe Ratio: GARCH (0.0691) > Historical (0.0645)
- Final cumulative return: GARCH (1.84) > Historical (1.66)

From a risk–return perspective, the GARCH portfolio is therefore superior, offering a higher return with a weekly volatility that is substantially comparable. The improvement in the Sharpe Ratio indicates that the excess return is not obtained by proportionally increasing risk, but thanks to a better allocation of securities within the class.

Analysis of stress weeks

Restricting the analysis to weeks of market stress, the superiority of the GARCH portfolio is not confirmed. In weeks characterized by negative shocks:

- the GARCH portfolio outperforms the historical counterpart only in 43.33% of cases;
- the strategy based on historical volatility shows greater defensive capability during adverse market phases.

This result is consistent with the literature on GARCH models: their high reactivity leads to an overestimation of volatility immediately after a shock, with a consequent aggressive reallocation of securities and greater short-term instability.

The analysis of the worst weekly drawdowns reinforces this interpretation. During extreme events (for example March 2020), the GARCH portfolio experiences significantly larger losses compared to the historical counterpart, which instead proves to be more resilient.

Comparison of portfolio composition

The analysis of portfolio composition at the date 2019-11-15 shows a relevant structural difference between the two approaches:

- a significant number of securities are selected exclusively by one of the two criteria.

This result indicates that the classification based on GARCH volatility is not a simple marginal reorganization of historical volatility, but produces a genuine recomposition of the investable universe. Consequently, the observed performance differences are not attributable to statistical noise, but to different allocation choices.

Overall assessment

Overall, the results show that:

- the classification based on GARCH volatility improves performance on average and over the long term;
- the strategy is, however, more vulnerable during stress phases, due to the high sensitivity of the model to volatility shocks;
- historical volatility offers greater defensive stability, but at the expense of cumulative return.

Aggressive class

Graphical comparison and cumulative performance

The graphical comparison between GARCH aggressive and historical volatility aggressive shows that the two cumulative return curves are initially very similar, but in the medium to long term the portfolio based on historical volatility tends to outperform the one constructed using GARCH.

In the invested capital chart (base 100), the historical strategy shows a slightly higher final growth, while the GARCH portfolio exhibits greater path irregularity, with wider fluctuations and greater sensitivity to market shocks.

This behavior suggests that, in the aggressive class, the high reactivity of the GARCH model amplifies portfolio volatility, without translating into a structural advantage in terms of cumulative return.

Quantitative performance comparison

Performance metrics confirm what is observed at the graphical level:

- Weekly average return: Historical (≈ 0.00290) > GARCH (≈ 0.00281)
- Sharpe Ratio: Historical (≈ 0.064) > GARCH (≈ 0.063)
- Final cumulative return: Historical (≈ 2.15) > GARCH (≈ 2.05)
- Max Drawdown: Historical (≈ -0.472) < GARCH (≈ -0.451)

Although the GARCH portfolio shows a slightly less severe maximum drawdown, this advantage is not sufficient to offset the lower cumulative return and the marginally lower Sharpe ratio. Overall, the strategy based on historical volatility appears more efficient in transforming risk into return for the aggressive class.

Analysis of stress weeks

The analysis of stress weeks provides a particularly relevant indication. In this subsample:

- the GARCH aggressive portfolio outperforms the historical counterpart in about 59% of the weeks.
- the GARCH strategy therefore shows a greater recovery capability during the most turbulent phases.

This result, apparently in contrast with overall performance, suggests that the GARCH model is able to react more quickly to volatility shocks in high-risk segments. However, this advantage manifests mainly in the short term and does not consolidate over the long term.

Analysis of the worst weekly drawdowns

The analysis of the five worst weekly crashes shows that, during extreme events (particularly March 2020), the GARCH portfolio suffers very deep losses, often comparable to or greater than those of the historical strategy. This highlights how the GARCH strategy remains strongly exposed to systemic shocks when applied to intrinsically volatile stocks. The high sensitivity of the model can therefore exacerbate losses at times of maximum turbulence.

Comparison of portfolio composition

The composition comparison shows a significant but not total overlap between the two portfolios. However, unlike the moderate class, this reorganization of the investable universe does not translate into an improvement in aggregate performance metrics.

Overall assessment

Overall, the results for the aggressive class indicate that:

- GARCH volatility improves portfolio reactivity during short-term stress phases, but this advantage is not sufficient to generate persistent long-term outperformance;
- the greater instability introduced by the model reduces overall risk–return efficiency.

Conclusions – Part 1

The analysis conducted in the first part of the work makes it possible to draw some general considerations on the use of forecast-based volatility measures, particularly those based on GARCH models, in the process of stock classification and portfolio construction. Overall, the results show that GARCH volatility contains additional information compared to historical volatility, but that this information does not translate into improved performance for all risk classes. The effectiveness of the model is in fact strongly dependent on the application context and on periods of market stress.

Main critical issues of the GARCH model

The evidence allows the identification of some critical issues in the use of GARCH as the sole classification criterion:

- Excessive reactivity to shocks: in the presence of extreme events, the model tends to overestimate future volatility.
- Instability during stress phases: the analysis of crisis weeks shows that GARCH portfolios suffer deeper drawdowns compared to historical counterparts, highlighting a weakness precisely during stress periods.

Considering these critical issues, the isolated use of GARCH volatility appears insufficient to obtain robust strategies across different market regimes. The evidence therefore suggests some directions for improvement:

- Combination of risk classes: the aggregation of conservative, moderate, and aggressive classes makes it possible to compensate for the weaknesses of the model in different contexts, exploiting the strengths of each segment.
- Introduction of dependence measures: the use of the var-covar matrix and clustering techniques makes it possible to integrate information on correlations among stocks, improving diversification and reducing systemic risk.

These considerations motivate the extensions developed in the next parts of the project.

Part 2

The second part of the work introduces an additional level of structure compared to simple volatility-based classification. After dividing stocks into risk classes based on GARCH volatility, the objective becomes the selection of a subset of stocks within each class, improving portfolio diversification and efficiency.

The var–covar selection uses the information contained in the variance–covariance matrix of returns estimated on the stocks belonging to the same risk class

. Two stocks with similar volatility may have:

- strongly correlated returns (poor diversification);
- weakly or negatively correlated returns (portfolio benefit).

Volatility-based classification alone does not capture this difference, whereas the var–covar matrix allows the dependence relationships to be modeled explicitly.

The var–covar matrix is therefore used not to directly optimize weights, but to:

- identify groups of stocks with similar behavior (clustering);
- select stocks belonging to different clusters;
- reduce concentration risk within each GARCH risk class.

Clustering is applied to the correlation matrix of returns within each GARCH risk class, after transformation into a distance matrix. In this way, stocks are grouped according to the similarity of their movements, allowing for a selection that favors structural diversification rather than the mere number of assets.

In this part of the work, the portfolios from the first part of the project are therefore compared to evaluate the effectiveness of the new model. In addition, there is also a comparison with a portfolio created using the best stocks per class in terms of returns, to assess whether the increase in complexity brings actual benefits to the strategy. In all three cases, the portfolios are equal weighted so that the quality of the selection alone can be evaluated.

Commentary on the results – Part 2

Conservative class

Performance results

Cumulative return

The strategy based on the var–covar matrix shows the highest cumulative return (1.3067), higher than both the GARCH baseline strategy (1.1921) and the Best 30 selection (1.0251). This result indicates that selection guided by the dependence structure among stocks makes it possible to improve capital growth over the long term.

Weekly average return

The weekly average return of the var–covar strategy (0.0019) is higher than that of the GARCH equal-weight portfolio (0.0017) and the Best 30 (0.0016), suggesting a better systematic contribution to returns, not driven by isolated single episodes.

Volatility

The weekly volatility of the var–covar portfolio (0.0207) is only slightly higher than that of the baseline strategy (0.0204) and comparable to that of the Best 30 (0.0209).

This indicates that the increase in returns is not accompanied by a significant increase in overall risk.

Sharpe Ratio

The Sharpe Ratio of the var–covar strategy (0.0616) is the highest among the three. This highlights an improvement in risk–return efficiency compared to both the equal-weight strategy and the selection based exclusively on past returns.

Max Drawdown

A particularly relevant result concerns the maximum drawdown.

The var–covar strategy records a drawdown of -0.2766 , significantly lower than that of the GARCH equal-weight strategy (-0.3354).

This suggests a greater ability to contain losses during periods of market stress.

Graphical analysis

Observation of the cumulative return chart shows that the trajectories of the var–covar strategy and the baseline are similar during regular market phases. However, during correction phases and periods of higher volatility, the var–covar portfolio tends to:

- experience less pronounced declines,
- recover more quickly,
- maintain a more regular profile over time.

The comparison shows that, within the conservative class estimated through GARCH, stock selection based on the var–covar matrix makes it possible to obtain a portfolio with:

- higher cumulative return,
- better Sharpe Ratio,
- significantly lower maximum drawdown, compared to the traditional equal-weight strategy.

These results provide empirical evidence in favor of using the correlation structure of returns as a selection criterion.

Moderate class

Performance results

Cumulative return

The strategy based on the var–covar matrix records a cumulative return of 2.2653, clearly higher than both the GARCH moderate equal-weight strategy (1.8435) and the Best 30 strategy (1.6610). This result highlights a significant improvement in long-term capital growth.

Weekly average return

The weekly average return of the var–covar strategy (0.0027) is higher than that of the baseline strategy (0.0024) and that of the Best 30 selection (0.0023).

The return differential appears stable over time, suggesting a systematic benefit derived from selection based on the dependence structure.

Volatility

The weekly volatility of the var–covar portfolio (0.0265) is aligned with that of the other strategies considered.

Therefore, the increase in returns is not accompanied by a significant increase in overall risk.

Sharpe Ratio

The Sharpe Ratio of the var–covar strategy (0.0792) is the highest among the three strategies. This result indicates a substantial improvement in risk–return efficiency compared to both the equal-weight strategy and the selection based on historical returns.

Max Drawdown

The maximum drawdown of the var–covar strategy is -0.3601 , slightly lower than that of the baseline strategy (-0.3640) and comparable to that of the Best 30 strategy (-0.3662).

This suggests that var–covar selection makes it possible to improve returns without significantly worsening exposure to extreme losses.

Graphical analysis

From the graphical comparison of cumulative returns, it emerges that the var–covar strategy shows a steeper and more regular trajectory compared to the equal-weight strategy.

In particular:

- during market expansion phases, the var–covar strategy consistently outperforms.
- during corrective phases, losses are comparable to those of the baseline, with no evidence of greater instability.

The comparison shows that, within the moderate class, stock selection based on the var–covar matrix makes it possible to obtain a portfolio with:

- significantly higher cumulative return,
- superior Sharpe Ratio,
- volatility and drawdown comparable to the reference strategies.

These results reinforce the empirical evidence in favor of using the correlation structure of returns as a selection criterion.

Aggressive class

Performance results

Cumulative return

The strategy based on the var-covar matrix records a cumulative return of 3.5890, clearly higher than both the GARCH aggressive equal-weight strategy (2.0542) and the Best 30 strategy (3.1666). This result indicates a strong improvement in long-term capital growth when selection takes into account the dependence structure among stocks.

Weekly average return

The weekly average return of the var-covar strategy (0.0036) is higher than that of the baseline strategy (0.0028) and slightly higher than that of the Best 30 selection (0.0034). The increase in average return suggests a systematic benefit derived from var-covar selection, particularly relevant in a high-risk class.

Volatility

The weekly volatility of the var-covar portfolio (0.0343) is slightly lower than that of the baseline strategy (0.0355) and comparable to that of the Best 30 (0.0338). This highlights that the increase in returns is not obtained through an increase in overall risk, but through a more efficient management of the risk structure.

Sharpe Ratio

The Sharpe Ratio of the var-covar strategy (0.0875) is the highest among the three strategies considered.

The improvement compared to the baseline strategy (0.0630) is particularly marked, indicating a significant increase in risk-return efficiency in the aggressive class.

Max Drawdown

The maximum drawdown of the var-covar strategy is -0.3573 , significantly lower than that of the GARCH aggressive equal-weight strategy (-0.4510).

This result is particularly relevant, as it indicates a greater ability to contain extreme losses in a class characterized by high volatility.

Graphical analysis

From the graphical comparison of cumulative returns, a clear separation emerges between the var-covar strategy and the baseline strategy.

In particular:

- during market expansion phases, the var-covar strategy shows faster growth.
- during correction phases, losses are less severe than those of the equal weight strategy.
- the overall return profile appears more regular and less subject to strong fluctuations.

The comparison highlights that, within the aggressive class, stock selection based on the var-covar matrix makes it possible to obtain a portfolio with:

- significantly higher cumulative return,
- markedly higher Sharpe Ratio,
- substantially lower maximum drawdown, compared to the GARCH aggressive equal-weight strategy.

Conclusions – Part 2

The introduction of criteria based on the var–covar matrix makes it possible to systematically improve performance compared to the baseline GARCH portfolio, especially in terms of cumulative return and Sharpe Ratio, with particularly evident effects in the moderate and aggressive classes. In these classes, where correlation among stocks plays a central role, var–covar selection allows for a reduction in risk concentration and a mitigation of deeper drawdowns.

The Best 30 selection, while improving returns in some cases, proves to be less robust from a risk–return perspective, showing a less stable advantage compared to the var–covar strategy. Overall, the results suggest that the true added value emerges when volatility estimation (GARCH) is complemented by a selection based on correlations among stocks in order to diversify the portfolio.

Part 3

Part 3 aims to assess whether the risk information used in the previous parts can translate into an advantage also at the aggregated portfolio level, while maintaining a coherent and comparable methodological framework. The choices made respond to three key criteria: simplicity, neutrality with respect to expected returns, and comparability of results.

Methodology

For the GARCH strategy, the adoption of equal-weight portfolios within each risk class is not a random choice, but a deliberate one:

- it eliminates dependence on estimates of expected returns, which are notoriously unstable and noisy;
- it allows the effect of volatility-based classification to be isolated, making the informational contribution of the GARCH model clearer compared to historical volatility.

In this way, any performance differences between portfolios are not attributable to allocation choices, but exclusively to the stock selection criterion based on risk.

For the market portfolio, a Minimum Variance optimization is used, consistently with the risk-based approach of the entire project. This choice is motivated by several reasons:

- the minimum variance portfolio represents the natural solution when the objective is the reduction of total risk;
- fair comparison with GARCH portfolios, since the portfolios constructed in the previous parts use exclusively volatility information, making the comparison with a benchmark that also optimizes only risk comparable;
- the absence of performance-based optimizations, eliminating the uncertainty of estimates related to expected returns.
- the minimum variance portfolio is for sure on the efficient frontier

Within this framework, any outperformance of the aggregated strategy relative to the benchmark does not stem from more aggressive allocation assumptions, but from a better organization of risk information, consistent with the overall design of the project.

Commentary on results and conclusions – Part 3

From the direct comparison between the two strategies, clear differences emerge.

In terms of weekly average return, the meta-strategy records a value of 0.270%, more than double the 0.118% of the minimum variance portfolio on the S&P 500. This difference on a weekly basis, translates into a strong cumulative effect over the long term, as highlighted by the final capital.

- Meta-strategy: +135.6% (final capital $\approx$ 235.6 on a base of 100)
- Minimum variance S&P 500: +71.4% (final capital $\approx$ 171.4 on a base of 100)

In relative terms, the meta-strategy generates a final capital higher by approximately +64 percentage points, significantly exceeding the benchmark. This result summarizes the combined effect of a higher weekly average return and a better ability to exploit market expansion phases.

From a risk perspective, the weekly volatility of the meta-strategy is 2.55%, higher than that of the benchmark (1.60%). However, this increase in volatility is accompanied by a more than proportional growth in returns, as shown by the weekly Sharpe ratio, which rises from 0.038 for the benchmark to 0.083 for the meta-strategy. This indicates greater efficiency in the use of the assumed risk.

Regarding maximum drawdown, the minimum variance portfolio shows a maximum loss of -21.8%, while the meta-strategy reaches a deeper drawdown (-33.1%). This result is consistent with the higher risk exposure and the absence of explicit defensive constraints. However, the analysis of invested capital shows that such drawdowns are recovered more quickly by the meta-strategy, which benefits more from market rebound phases.

The set of metrics shows how the minimum variance portfolio effectively achieve the function of risk reduction, but at the cost of a significant compression of expected returns. Contrarily, the meta-strategy, while accepting greater return variability and more pronounced drawdowns, manages to achieve a markedly higher remuneration of risk, as demonstrated by the Sharpe ratio (meta-strategy Sharpe ratio $\approx$ 0.083, minimum variance Sharpe ratio $\approx$ 0.038) and by cumulative returns.

These findings suggest that, within a methodologically coherent comparison framework free of expected return estimates, an approach based on volatility and diversification across risk classes may be structurally more efficient than a purely defensive minimum variance strategy.

Overall, the results suggest that volatility, when used in a structured manner and integrated with information on dependence among assets, represents an effective informational variable for the construction of robust portfolios. However, it clearly emerges that excessive reactivity of conditional volatility models and the absence of constraints on exposure can amplify drawdowns during crisis phases, indicating possible future extensions.